

Wonderland SOC

Incident Investigation Notes

Finding name: Creation of Shadow Copy

Analyst assigned: Elizabeth Watt

Description of System(s)	
Type (Laptop/Desktop/Server/VM/etc.)	On Prem Windows 192.168.0.0/24 host/laptop mvalitus I with the user mvalitus 192.168.70.167 (downloaded suspicious file)
System Operating System	Windows
System Name	Frothly Active Directory Server Labrador
System IP Address	192.168.70.150
Other hosts involved: Hostname/IP address	mvalitus I

User information (as applicable)	
User(s) involved in incident	pcerf
Other user(s) alias involved	mvalitus

MITRE ATT&CK info	
Tactic / Technique - ID	Credential Access TA0006 / Technique ID T1003.003 Sub technique: Remote Windows Management Technique ID: T1021.006 Proxy: Domain Fronting/Technique ID: T1090.004

Wonderland SOC

Incident Investigation Notes

Investigation Details

High Urgency notable event Creation of Shadow copy
Shadow copy a snapshot of a volume that duplicates all of the data held on that volume at an instant in time (even if the data is in use). Shadow copy was created using either vssadmin or wmic according to the event description. Frothly Active Directory Server, Labrador appears to have been copied

Labrador:
Large volume of outbound web traffic
Remote PowerShell launches
Registry Autoruns being added

pcerf@froth.ly, excessive failed logins, category:technical, priority: high

Shadow copy was deleted shortly after it was created which is suspicious

print spooler service uses wscript to execute a node.js file, then creates and deletes a volume shadow copy, then moves a bunch of files around on the domain controller, user pcerf is mapping a network drive to this domain controller, copying all those files; mounting a network drive; moving all those files out to the network drive

Remote powershell possible lateral movement

outbound traffic to 152.195.19.97

Additional Artifacts

Consulted SME to discuss findings, she suggested to keep searching and it could possibly be a true positive. The search confirmed the finding vssadmin.exe executed twice, less than 3 minutes apart. SME suggested that I confirm with the IT department to make sure that the department didn't order it.

After I verified with Grace, Frothly's IT department didn't authorize this and it wasn't their employee Peat Cerf. Determined this is a true positive.

wscript.exe process and the file "hefeweizen_tips.js

file hefeweizen_tips.js being downloaded to mvalitus l.froth.ly at 18:31 on 8/18/2020

ip address 46.101.247.84 this website may be the source of the javascript under investigation

suspicious website <https://dunkel.hefeweizen.azureedge.net/index.html>

Final Analysis

Wonderland SOC

Incident Investigation Notes

SPL Searches

3.1 Windows Event logs search for vssadmin.exe

```
index=main sourcetype=xmlwineventlog EventCode=1 dest=labrador process_name=vssadmin.exe  
| table _time user process process_exec parent_process  
| sort _time +
```

3.2 Search for Peat Cerf credentials on Labrador

```
| tstats summariesonly=true count from datamodel=Endpoint.Processes where  
Processes.parent_process_name="spoolsv.exe" Processes.dest="labrador.froth.ly" Processes.user="pcerf"  
groupby _time span=1s Processes.parent_process Processes.process
```

3.3 search across our data to see if there are any other instances of Windows Script Host being used to run suspicious processes or functions

```
| tstats summariesonly=true count from datamodel=Endpoint.Processes where  
Processes.process_name="wscript.exe" groupby _time span=1s Processes.process  
Processes.process_name Processes.parent_process Processes.dest Processes.user
```

3.4 search endpoint data model hefeweizen_tips.js file

```
| tstats summariesonly=true count values(Filesystem.file_size) AS file_size from  
datamodel=Endpoint.Filesystem where Filesystem.file_name="*hefeweizen_tips.js*" groupby _time span=1s  
Filesystem.file_name Filesystem.file_path  
| drop_dm_object_name("Filesystem")  
| table _time file_name file_path  
| sort + _time
```

3.5 Web Data Model

```
| from datamodel:"Web"."Web"  
| search (url="*hefeweizen_tips.js*")  
| head 100
```

3.6

```
transaction sourcetype=stream:dns dest_ip="192.168.70.150" answer="152.195.19.97" record_type=A  
| table timestamp hostname{}
```

3.6.b

```
index=main host=labrador dest_ip=46.101.247.84 AND *hefeweizen  
| sort _time
```